

HEALTHCONNECT CLINIC EXPERIENCE LAB

Week 5 - Data Analytics Track

Initial Analytics Report, Business Questions, KPIs, Insights & Recommendations

Prepared from the Week 4 Initial Analysis, HealthConnect Power BI dashboard, and official Week 5 assignment brief

Executive. Summary

The Week 5 Data Analytics work moves HealthConnect from Week 4 planning into practical implementation. The analysis uses the 5,000-record HealthConnect appointment dataset and the Power BI dashboard to assess no-show patterns, calculate the five proposed KPIs, interpret the findings and translate them into operational recommendations.

The main finding is that no-shows are strongly associated with actionable booking and behavioural characteristics. Overall no-show rate is 48.5%. The largest displayed gradient is booking lead time, ranging from 24.8% for 0-3 day bookings to 60.5% for 30+ day bookings. Patients with a prior no-show have a 55.4% current no-show rate versus 43.5% among patients without a prior no-show. Reminder channel and distance also show meaningful differences, while broad demographic differences are comparatively smaller.

The dashboard also reports a 53.7% Lost Slot Rate, indicating a major capacity-utilisation problem. The recommended response is targeted: strengthen confirmation for long-lead bookings, flag prior no-show patients, optimise reminders, prioritise manual outreach and explore access alternatives for distant patients. These are evidence-informed recommendations and should not be interpreted as proof of causation.

1. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 fictional, anonymized appointment records across 1,696 unique patients, averaging approximately 2.9 appointments per patient. Records cover 18 variables across demographics, appointment details, booking information, patient history, reminders, access measures and outcomes.

Variable group	Fields
Patient demographics	Gender, age, age group
Appointment details	Appointment type, appointment day, appointment time
Booking information	Booking date, appointment date, booking lead days
Patient history	Previous appointments, previous no-shows
Reminder, access and outcome	Reminder sent, reminder channel, distance to clinic, waiting time, appointment outcome

Appointment dates span 1 January 2025 to 30 June 2026, with bookings placed as early as 7 November 2024.

Outcome	Count	%
Attended	2,314	46.3%
No-Show	2,423	48.5%
Cancelled	263	5.3%

2. Week 4 to Week 5 Transition

Week 4 defined the problem, business questions, potential KPIs, data-quality observations and the proposed Power BI approach. Week 5 operationalises those decisions through data preparation, exploratory analysis, KPI calculation, dashboard interpretation and business recommendations. The project therefore continues the same HealthConnect problem rather than introducing a new dataset or business problem.

Week 4 foundation	Week 5 implementation
Defined the missed-appointment problem	Measured the scale and patterns of missed appointments
Identified five potential KPIs	Calculated/interpreted the five KPIs
Assessed data quality	Documented preparation findings and treatment decisions
Planned dashboard analysis	Used dashboard evidence to support findings
Identified business questions	Answered the questions using EDA, KPIs and recommendations

3. Business Questions

1. Which patient, booking and appointment characteristics are most strongly associated with no-shows?
2. Does sending an appointment reminder, and the channel used, measurably influence attendance?
3. How much appointment capacity is lost to no-shows and cancellations, and where does this loss concentrate?
4. Are patients with a prior history of missed appointments more likely to miss future appointments, and can this history be used to flag higher-risk bookings?
5. How do distance to the clinic and booking lead time relate to attendance behaviour?

4. Dataset Preparation and Data Quality

The preparation stage focused on data completeness, uniqueness, consistency and the business meaning of important fields. The original dataset is preserved; preparation decisions are documented separately so downstream analysis can be reproduced and reviewed.

Check	Finding	Assessment	Treatment/decision	Impact on analysis
Reminder channel missingness	1,366 records (27.3%)	Expected; exactly matches reminder sent = No	Treat as a meaningful no-reminder state; do not impute	Preserves the meaning of no reminder
Distance missingness	90 records (1.8%)	Minor; appears randomly distributed across outcomes	Document decision; use known values for distance comparisons unless an explicit imputation rule is adopted	Avoids inventing distance values
Waiting time missingness	60 records (1.2%)	Minor, but field also has a conceptual anomaly	Exclude from no-show driver interpretation; if used, restrict to appropriate attended records	Prevents a misleading causal interpretation
Duplicate appointment IDs	None found	Unique across 5,000 records	No duplicate removal required	Supports appointment-level grain
Fully duplicated rows	None found	No fully duplicated records	No duplicate removal required	Preserves source records
Historical consistency	0 violations	Previous no-shows never exceed previous appointments	Retain	Supports use of history fields
Date consistency	No inconsistencies	Appointment date never precedes booking date; lead days reconcile	Retain	Supports lead-time analysis
Categorical values	No stray categories	Values align with Data Dictionary	Retain documented categories	Reduces category ambiguity

4.1 Waiting-time anomaly and treatment

The Week 4 assessment found `waiting_time_minutes` populated for 2,386 of 2,423 no-shows and 261 of 263 cancellations, with averages around 24 minutes across outcome groups. Because patients who did not attend could not have recorded an actual clinic waiting time, the field is likely simulated or pre-assigned. It is therefore not treated as a causal no-show driver and is flagged for downstream modelling work.

4.2 Preparation conclusion

The dataset is sufficiently structured for Week 5 exploratory analysis and KPI development. The main preparation requirement is transparent handling of missing values and careful treatment of variables whose business meaning is limited.

5. Exploratory Data Analysis

Analysis area	Observed result	Interpretation	Business implication
---------------	-----------------	----------------	----------------------

Analysis area	Observed result	Interpretation	Business implication
Overall outcome	48.5% no-show rate	Nearly half of scheduled appointments are missed	Attendance and capacity utilisation are priority issues
Appointment type	Follow-up Diagnostic Test 51.23%; Specialist Consultation 49.75%; General Consultation 47.44%	Some services show higher no-show rates	Targeted service-level interventions may be appropriate
Reminder channel	None 51.39%; WhatsApp 49.77%; Email 48.41%; SMS 45.75%	No-reminder is highest; SMS is lowest	Reminder process is an actionable lever to test
Prior no-show history	55.4% vs 43.5%	Previous behaviour is associated with future attendance risk	Supports targeted risk flagging
Lead time	0-3 days 24.8%; 4-14 32.6%; 15-30 43.2%; 30+ 60.5%	Strong increasing gradient	Long-lead bookings need earlier confirmation
Distance	<5 km 46.40%; 5-15 km 47.26%; 15+ km 54.11%	Farthest group has higher observed no-show rate	Access/travel burden may be relevant
Gender	Male 48.71%; Female 48.43%; Prefer not to say 43.52%	Differences are comparatively small	Demographic targeting is less immediately actionable
Age group	45.12% to 50.75%	Range is narrower than lead-time range	Operational variables appear stronger first-line targets

6. KPI Development, Business Questions and Interpretation

The five KPIs below were proposed in Week 4 and are now operationalised in Week 5. Each KPI is explicitly connected to a business question, a management need, the calculation logic, the observed result and its interpretation.

KPI	Question answered	Why relevant	Calculation	Current result	Interpretation
No-Show Rate	Q1 and Q3	Measures the scale of missed appointments and supports segmentation across patient, booking and appointment characteristics.	No-Show appointments / Total appointments	48.5%	Nearly half of scheduled appointments are missed. This is the baseline measure for assessing improvement and locating higher-risk segments.
Reminder Effectiveness	Q2	Tests whether reminder status/channel is associated with different attendance outcomes and evaluates a controllable operational lever.	Compare no-show/attendance rates for reminder sent vs not sent and by channel.	None 51.39%; WhatsApp 49.77%; Email 48.41%; SMS 45.75%	SMS has the lowest displayed no-show rate and no-reminder the highest. The finding supports reminder optimisation but does not prove causation.
Lost Slot Rate	Q3	Shows the proportion of scheduled capacity not converted into a completed visit.	(No-Show + Cancelled) / Total appointments	53.7%	More than half of scheduled capacity is reported as lost through no-shows or cancellations, making capacity recovery a major operational priority.
Repeat No-Show Rate	Q4	Tests whether previous missed appointments are associated with higher future no-show risk.	No-show rate where previous_no_shows > 0, compared with = 0.	55.4% vs 43.5%	The 11.9 percentage-point gap makes prior attendance history a useful operational risk signal and a candidate predictive feature.
Distance / Lead-Time Attendance	Q5	Tests whether access burden and booking behaviour are associated with attendance.	Compare no-show/attendance rates across distance and lead-time bands.	Lead time 24.8%-60.5%; distance 46.4%-54.1%	Lead time shows the strongest observed gradient. Distance also shows a higher rate among the farthest known-distance group.

6.1 KPI-to-question mapping

Business question	Primary KPI(s)	Current analytical answer
Q1: What characteristics are associated with no-shows?	No-Show Rate (segmented)	Lead time, prior history and some appointment types show clearer differences.
Q2: Do reminders influence attendance?	Reminder Effectiveness	Reminder groups differ; SMS has the lowest displayed no-show rate.
Q3: How much capacity is lost?	Lost Slot Rate	53.7% of scheduled capacity is reported as lost

Business question	Primary KPI(s)	Current analytical answer
		to no-shows/cancellations.
Q4: Do prior no-shows matter?	Repeat No-Show Rate	55.4% vs 43.5%, a substantial observed difference.
Q5: Do distance and lead time matter?	Distance / Lead-Time Attendance	Lead time shows the widest gradient; distance also varies by band.

7. DAX Measures

The following measures are suitable starting points for the Power BI model. Replace the table name if your model uses a different name.

Total Appointments

```
Total Appointments =
COUNTROWS('Appointments')
```

No-Show Appointments

```
No-Show Appointments =
CALCULATE([Total Appointments], 'Appointments'[appointment_outcome] = "No-Show")
```

No-Show Rate

```
No-Show Rate =
DIVIDE([No-Show Appointments], [Total Appointments], 0)
```

Cancelled Appointments

```
Cancelled Appointments =
CALCULATE([Total Appointments], 'Appointments'[appointment_outcome] = "Cancelled")
```

Lost Slot Rate

```
Lost Slot Rate =
DIVIDE([No-Show Appointments] + [Cancelled Appointments], [Total Appointments], 0)
```

Repeat No-Show Rate

```
Repeat No-Show Rate =
DIVIDE(CALCULATE([Total Appointments], 'Appointments'[previous_no_shows] > 0,
'Appointments'[appointment_outcome] = "No-Show"), CALCULATE([Total Appointments],
'Appointments'[previous_no_shows] > 0), 0)
```

Clean-History No-Show Rate

```
Clean-History No-Show Rate =
DIVIDE(CALCULATE([Total Appointments], 'Appointments'[previous_no_shows] = 0,
'Appointments'[appointment_outcome] = "No-Show"), CALCULATE([Total Appointments],
'Appointments'[previous_no_shows] = 0), 0)
```

No-Reminder Rate

```
No-Reminder Rate =
DIVIDE(CALCULATE([Total Appointments], 'Appointments'[reminder_sent] = "No"), [Total Appointments], 0)
```

No-Show Rate - Reminder Sent

```
No-Show Rate - Reminder Sent =
CALCULATE([No-Show Rate], 'Appointments'[reminder_sent] = "Yes")
```

For distance and lead-time analysis, use categorical bands as dimensions and reuse the No-Show Rate measure so that the calculation responds to Power BI filter context.

8. Dashboard / Analytical Visualisation

The current Power BI dashboard provides the initial analytical visual report required for Week 5. It contains KPI cards, segmentation charts and a key-insights/recommendations section. The most decision-useful visuals are the lead-time chart, prior-no-show comparison, reminder-channel comparison, distance comparison, appointment-type comparison and the two primary KPI cards.

Visual	Purpose	Evidence	Decision use
No-Show Rate card	Shows baseline problem size	48.5%	Establishes urgency
Lost Slot Rate card	Shows capacity loss	53.7%	Frames the issue operationally
Lead-time bar chart	Shows strongest gradient	24.8%-60.5%	Prioritises confirmation
Prior-no-show chart	Shows behavioural difference	55.4% vs 43.5%	Supports risk flagging
Reminder-channel chart	Compares reminder patterns	SMS 45.75% vs None 51.39%	Supports reminder testing
Distance chart	Shows access gradient	46.4%-54.1%	Supports access interventions
Appointment-type chart	Shows service differences	51.23%-47.44%	Supports targeted workflows

9. Key Business Insights

Insight 1 - Booking lead time is the strongest observed pattern. No-show rate increases from 24.8% for 0-3 day bookings to 60.5% for 30+ day bookings, making lead time the strongest displayed operational segmentation signal.

Insight 2 - Previous no-shows are a meaningful behavioural signal. Patients with a prior no-show have a 55.4% current no-show rate versus 43.5% among those without a prior no-show, supporting a targeted risk flag.

Insight 3 - Reminder practices show a modest actionable association. No-reminder appointments have the highest displayed no-show rate at 51.39%, while SMS has the lowest at 45.75%.

Insight 4 - Distance appears relevant to attendance. The observed rate increases from 46.40% for patients under 5 km to 54.11% for patients 15 km or more away among known-distance bands.

Insight 5 - Capacity loss is substantial. The 53.7% Lost Slot Rate indicates that a large share of scheduled capacity does not produce a completed appointment.

Insight 6 - Appointment type differences justify targeted review. Follow-up Diagnostic Test has the highest displayed no-show rate at 51.23%, followed by Specialist Consultation at 49.75% and General Consultation at 47.44%.

Insight 7 - Demographic differences are comparatively smaller. Gender rates cluster around the overall rate and age-group rates range from 45.12% to 50.75%, suggesting more actionable first-line levers elsewhere.

10. Business Recommendations

Recommendation	Evidence	Action	Expected benefit	Priority
Confirm long-lead bookings	30+ day no-show rate = 60.5%	Add stronger confirmation checkpoints for bookings made 15+ days ahead.	Reduce avoidable no-shows and protect future capacity.	High
Flag prior no-show patients	55.4% vs 43.5%	Create a simple operational flag for previous_no_shows > 0 and prioritise follow-up.	Target staff effort toward higher observed risk.	High
Strengthen SMS reminders	SMS 45.75%; no reminder 51.39%	Use SMS as default/preferred where feasible and monitor outcomes.	Potentially improve attendance through better reminder coverage.	High
Prioritise confirmation calls	Risk concentrated in identifiable segments	Use lead time, prior history, distance and appointment type to prioritise calls.	Use staff time more efficiently.	Medium
Explore access alternatives	15 km+ rate = 54.11%	Assess	Address possible travel	Medium

Recommendation	Evidence	Action	Expected benefit	Priority
		telehealth/alternative access where clinically and operationally appropriate.	barriers.	
Review higher-risk appointment types	Follow-up Diagnostic Test = 51.23%	Review patient preparation and confirmation workflow for higher-no-show service types.	Target the workflow where observed risk is higher.	Medium
Monitor Lost Slot Rate	53.7%	Track Lost Slot Rate by month and key segments after interventions.	Measure capacity recovery over time.	High

11. Assumptions, Limitations, Risks and Dependencies

11.1 Assumptions

- The dataset is fictional and synthetic; findings are for learning and may not mirror real clinic behaviour.
- Reminder-channel nulls represent no reminder sent because the missing count exactly matches reminder_sent = No.
- One record represents one appointment, consistent with the Week 4 quality assessment.

11.2 Limitations

- The analysis is observational and identifies associations rather than causation.
- waiting_time_minutes is not treated as a reliable attendance-linked explanatory variable.
- distance_to_clinic_km has 90 missing records (1.8%), so distance comparisons apply to known-distance records unless a documented imputation rule is adopted.
- There is no direct cost field; any financial estimate would require an assumed cost per slot.
- Cancelled and No-Show are different behaviours and should remain separate in behavioural analysis, although both contribute to Lost Slot Rate.
- Predictive usefulness of candidate variables should be validated by Data Science.

11.3 Risks

- Causal overinterpretation may lead to inappropriate decisions.
- Using the waiting-time field as a predictor could introduce misleading signal.
- Changing outcome definitions without re-running KPIs could create inconsistent reporting.
- Interventions may not produce the expected effect; results should be monitored after implementation.

11.4 Dependencies

Dependency	Why it matters	Action
HealthConnect Data Dictionary	Defines field meaning and categories.	Use as the source of truth.
Stable outcome definitions	KPI calculations depend on consistent labels.	Do not change labels without revalidation.
Data Science modelling	Uses the same dataset and candidate features.	Share findings and data-quality caveats.
Dashboard/model consistency	Results should agree across project outputs.	Reuse documented measure logic and filter rules.

12. Week 5 Project Summary

Required point	Week 5 outcome
Planned	Move from Week 4 planning into data preparation, EDA, KPI calculation, dashboard analysis and recommendations.
Completed	Data-quality findings documented; five KPIs interpreted; dashboard evidence analysed; business insights and recommendations

Required point	Week 5 outcome
	developed.
Key findings	Lead time is the strongest observed pattern; prior no-show history is a meaningful signal; reminders and distance show actionable associations; capacity loss is high.
Major challenges	Handling missing values transparently and excluding waiting time from causal no-show interpretation.
Important decisions	Treat reminder-channel nulls as no reminder; keep Cancelled separate from No-Show for behaviour; flag waiting-time anomaly.
Cross-track collaboration	A separate Data Science collaboration report documents the handoff and dependency.
Remaining work	Further segmentation, predictive validation, intervention monitoring and dashboard refinement.
Week 6 focus	Compare descriptive findings with model performance and strengthen decision-focused reporting.

13. Conclusion

The Week 5 Data Analytics work establishes a clearer evidence base for HealthConnect's no-show problem. A 48.5% no-show rate and 53.7% Lost Slot Rate indicate a substantial operational challenge. The strongest observed patterns are concentrated in actionable variables, particularly booking lead time and previous no-show history, with reminders, distance and appointment type providing additional segmentation opportunities.

The recommended response is targeted and testable: strengthen long-lead confirmation, flag repeat-risk patients, optimise reminder practice, prioritise higher-risk follow-up and assess access alternatives. Subsequent project stages should validate these associations through predictive modelling and monitored operational interventions.

14. Source Documents

- AnalystLab Africa - Week 5 Experience Lab Assignment.
- HealthConnect Clinic Experience Lab - Week 4 Initial Analysis Document (Data Analytics Track).
- HealthConnect Clinic No-Show Analysis Dashboard - Power BI export.